One of the two molecules of UQ produced diffuses to the matrix side of mitochondria where it is reduced to form UQH_2. The so formed UQH_2 diffuses to its oxidation site and joins the pool of UQH_2 from complexes I and II. During this process, 2H is released into the intermembrane space.

Cytochrome c oxidase: Cytochrome c oxidase of Complex IV catalyzes the one electron oxidation of four consecutive reduced cytochrome c molecules and the concomitant four electron reduction of one oxygen molecule. Cytochrome c oxidase contains two cytochromes (a and b). Cyt a is paired with copper atom, (CuA) and Cyt a3 is paired with a different copper atom, (CuB). Each of the reduced Cytc molecules donates two electrons, one at a time to CuA. The electrons are further transferred to cyt a, cyt a_3, CuB. The transfer of four electrons from cyt c converts oxygen and four protons to two molecule of water (Fig.8.8).

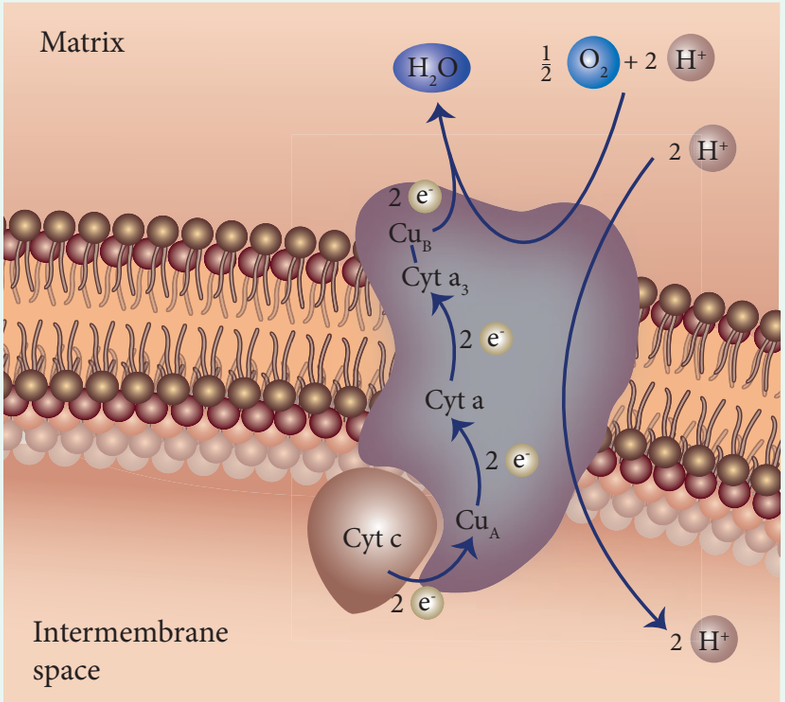


Figure 8.8 Overview of reactions in Complex IV

Complex No.	Complex Name	Polypeptides	Prosthetic groups
Complex I	NADH dehyrogenase	Around 46	FMN, Fe-S Centres
Complex II	Succinate dehydrogense	2 Fe-S Proteins	FAD, Fe-S Centres
Complex III	Cytochrome b c_1 complex	2 Cytochromes 1 Fe-S protein 6-8 other proteins	b and c type hemes (cyt b_1, b_H, c_1) Fe-S centres
Complex IV	Cytochrome oxidase	Around 13	a type hemes (cyt a,a_3) 2 Cu

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